

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY**Chandubhai S Patel Institute of Technology**4th Semester B.Tech. (IT)

University Exam Nov-Dec 2011

MA-206 Probability, Statistics and Numerical Analysis

Date: 06/12/'11

Time: 10:00am to 01:00pm Maximum Marks: 70

Instruction:

1. This question paper contains two sections **Section-I** and **Section-II**.
2. **Section-I** carries 35 and **Section-II** carries 35 marks.
3. Section wise separate answer book to be used.
4. Figure to the right indicates the distribution of full marks for the respective question.

Note: Normal distribution table is given at the end of the paper.**Section-I****Q-1 Attempt any two.****[10]**

- (a) Write a short note on Pie-chart.
- (b) Compute the Arithmetic Mean for the following data.

Marks	0-10	10-20	20-30	30-40	40-50
No. of students	2	18	30	17	3

- (c) Draw a Histogram and draw your conclusion for the data given in Q-1 (b).

Q-2 Attempt any three.**[15]**

- (a) Find the correlation coefficient between the sales and expenses of the following 10 firms using Karl Pearson's Product moment method. (Figures are in thousands)

Firms	1	2	3	4	5	6	7	8	9	10
Sales	50	50	55	60	65	65	65	60	60	50
Expenses	11	13	14	16	16	15	15	14	13	13

- (b) Two judges in a beauty contest rank the 12 contestants as follows

X	1	2	3	4	5	6	7	8	9	10	11	12
Y	12	9	6	10	3	5	4	7	8	2	11	1

What degree of agreement is there between the judges? Use Spearman's Rank Correlation Method.

- (c) Find the Standard Deviation of 28, 26, 22, 30, 20, 14, 18, 24 and 20.
- (d) A card is Drawn at random from a pack of 52 cards. What is the probability that the card is a spade or a king?

Q-3 From the following data, obtain (a) the regression line y on x and
(b) the regression line x on y .

[10]

x	1	2	3	4	5
y	9	11	5	8	7

==OR==

Q-3 A sample of 100 dry battery cells tested and found that average life of 12 hours and a standard deviation 3 hours. Assuming the data to be Normally distributed, what percentage of battery cells are expected to have life

- (i) More than 15 hours.
- (ii) Less than 6 hours.
- (iii) Between 10 and 14 hours?

Section-II

Q.1 Attempt any Three

[12]

- (i) Use Newton-Raphson method to find the root of $x^3 + x^2 + x - 100 = 0$, correct to three decimal places, considering $x_0 = 4$ is an initial root.
- (ii) Explain round off and truncation errors with suitable examples, and give difference between them.
- (iii) Using Newton's divided difference formula, calculate the value of $f(6)$ from the following data:

x	1	2	7	8
$f(x)$	1	5	5	4

- (iv) Evaluate $\int_0^1 x dx$, by Gaussian Integration.

- (v) Using Newton's backward interpolation formula, estimate the premium for policy maturing at the age of 63, the following table of half-yearly premium for policies maturing at different ages.

Age	45	50	55	60	65
Premium(in Rs)	114.84	96.16	83.32	74.48	68.48

Q.2 Attempt any Four

[12]

- (i) Evaluate $\int_0^2 e^{-x^2} dx$ by Simpson's $\frac{1}{3}$ rule with $h = 0.5$
- (ii) Show that $\Delta = e^{hD} - 1$.
- (iii) Using Lagrange's Formula find a polynomial of degree three which fits into the data below:

x	-1	0	1	3
y	2	1	0	-1

- (iv) Add and subtract the following floating point numbers:
(i) .4546E3 and .5433E7 (ii) 0.999658E-3 and 0.994576E-3
- (v) Using appropriate interpolation formula, calculate $f(5)$ from the following data:

x	4	6	8	10
y	1	3	8	16

- (vi) Define the following terms with appropriate examples:
(i) Relative error (ii) Percentage error and (iii) Absolute error

- Q.3(a) Find $y(1)$ from the differential equation $\frac{dy}{dx} = y + x, y(0) = 1$ with $h = 0.1$, using Euler's method. [5]

- Q.3(b) Using Runge-Kutta method of fourth order to solve the following differential equation, in the interval $[0, 0.4]$

$$\frac{dy}{dx} = \frac{y+x}{y-x}, y(0) = 1 \text{ and } h = 0.2. \quad [6]$$

OR

- Q.3(b) Fit a straight line for the data:

[6]

x	1	2	3	4	5
y	14	27	40	55	68